

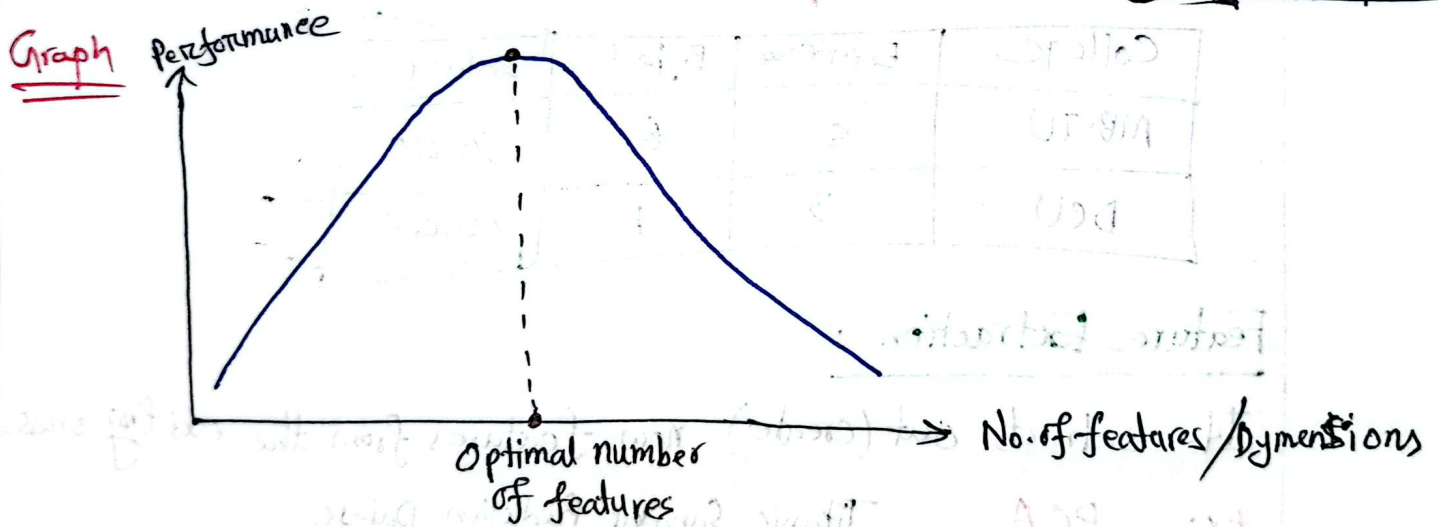
Dimensionality Reduction Feature Select

Curse of Dimensionality $\rightarrow$ If no. of features increased, it hurts model performance.

Ex: Salary Prediction Data

- Irrelevant features
- redundant features.

College/University	Project	Experience	weight	height (feet)	height (meter)	location
Buet	5	2	50	5.7	1.3	Ctg
DU	3	5	55	6.1	1.4	Dhaka
MBSTU	4	7	53	5.6	1.25	Craipur
...						...



Curse of Dimension

Using some
Technique

Dimensionality Reduction

(ex.)

feature selection

Feature Selection:

Feature selection is a way of selecting the subset of the most relevant features from the original features set by removing the redundant, irrelevant, or noisy features.

Salary Prediction Dataset

College	Experience	Weight	Project	Salary
MBSTU	5	58	6	90,000
DCU	3	62	1	60,000



College	Experience	Project	Salary
MBSTU	5	6	90,000
DCU	3	1	60,000

Feature Extraction:

It extracts out (create) new features from the existing ones.

Ex: PCA

Titanic Survival Prediction Dataset

Category	Gender	Age	Survive (0/1)
2	Male	53	0
3	Female	30	1
1	Male	8	1

↓ Gender + Age

Category	Gender	Age	IsChildOrFemale	Survive (0/1)
2	Male	53	0	0
3	Female	30	1	1
1	male	8	1	1

Types of Features selection

- Filter method
- Wrapper method
- Embedded Method.

Filter Method

↳ Checks the relevance of the feature with the output variable.

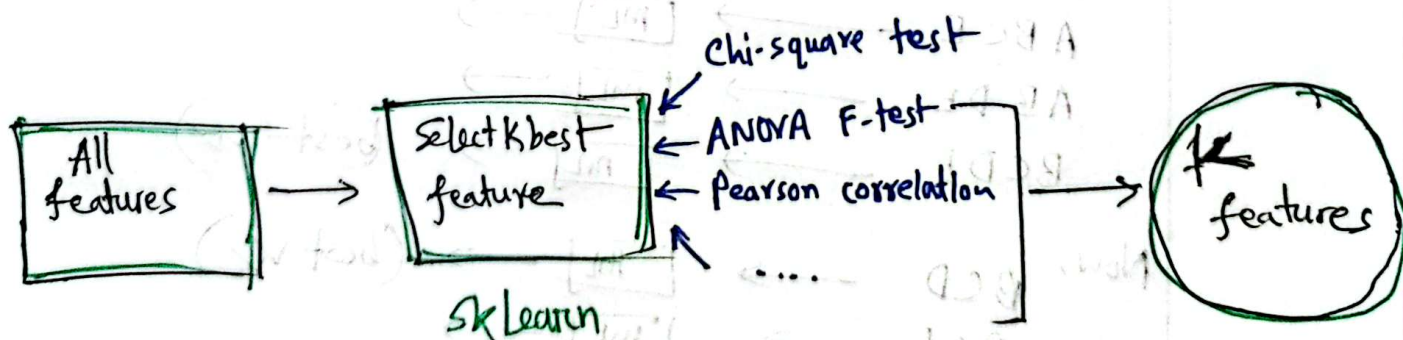
- Pearson Correlation coefficient
- Chi-square Test
- Spearman's Rank Correlation coefficient

Experience	Height (cm)	Salary
5	150	90,000
2	170	60,000
4	180	80,000

Relation

No relation

So,



App

Filter Method $\rightarrow$ before applying ML algorithm.

Wrapper Method $\rightarrow$ along with ML algorithm

Wrapper Method $\rightarrow$ Model/ML Algorithm Agnostic

- Forward Selection
- Backward Elimination

We can apply Wrapper method to any ML Algorithm.

Performance

Forward Selection

A	B	C	D	E	Op

A $\rightarrow$ ML $\rightarrow$

B $\rightarrow$ ML $\rightarrow$ ✓ (best)

C $\rightarrow$ ML $\rightarrow$

D $\rightarrow$ ML $\rightarrow$

E $\rightarrow$ ML $\rightarrow$

Now,

BA $\rightarrow$ ML $\rightarrow$

BC $\rightarrow$ ML $\rightarrow$

BD $\rightarrow$ ML $\rightarrow$ ✓ (Best)

BE $\rightarrow$ ML $\rightarrow$

So, if we need 2 feature we take B and D.

If we need 3 feature, apply same procedure.

Backward Elimination

Performance

ABCDE $\rightarrow$ ML $\rightarrow$

ABCD $\rightarrow$ ML $\rightarrow$

ABCE $\rightarrow$ ML $\rightarrow$

ABDE $\rightarrow$ ML $\rightarrow$

BCDE $\rightarrow$ ML $\rightarrow$ (best ✓)

Now,

BCD $\rightarrow$ ML $\rightarrow$ (best ✓)

BCE $\rightarrow$ ML $\rightarrow$

CDE $\rightarrow$ ML $\rightarrow$

BDE $\rightarrow$ ML $\rightarrow$

So, if we need 3 features, we take B, C and D.

Too easy
+
But not very powerful

Filter + wrapper

Too complex
+
But powerful

Embedded Method

- It combines the qualities of filter and wrapper methods.
- L1 regularisation, Decision trees, RandomForest etc.

Let's Code